

Cross-phase modulation in optical fibers

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When pulse pairs copropagate in a fiber, each pulse chirps the other through cross-phase modulation (XPM). In particular, XPM causes spectral broadening of signal pulses during Raman or parametric optical pulse amplification. We have studied XPM experimentally in the fiber Raman amplification soliton laser. We show that experimentally observed spectra can be explained by a simple model that concentrates on the phase change as a function of pulse walk-off.

Raman or parametric optical pulse amplification in fibers involves copropagation of a pump pulse and a weaker signal pulse. The nonlinear index change induced by the pump ($\delta n = n_2 I_p$) can phase modulate and thereby spectrally broaden the signal.¹ This cross-phase modulation (XPM) is largely responsible for the broad spectral width observed from fiber Raman soliton lasers^{2,3} and has also been demonstrated to be a strong effect in bulk glasses.⁴

The spectral features and broadening resulting from XPM depend on the walk-off between the pump and signal pulses. These spectral features can be confusing and complicated, but we show that they can be understood both qualitatively and quantitatively by concentrating on the phase change as a function of walk-off. XPM is most pronounced when the pump and signal are of comparable pulse widths and when they track each other. The phase change $\Delta\phi$ induced on the signal is proportional to the pump intensity, and the signal spectrum [Fig. 1(a)] looks like the one obtained from self-phase modulation (SPM). Nevertheless, the phase shift for the signal is twice that experienced by the pump because twice as many $\chi^{(3)}$ terms contribute to XPM as do to SPM.¹

The opposite extreme occurs when the phase shift is uniform over the width of the signal pulse. This may happen in the absence of pump depletion or spreading if the pump walks completely through the signal or if the signal is much narrower than the pump and precisely tracks the pump. XPM is canceled in this limit, and the original spectral width of the signal (much narrower than any shown in Fig. 1) results.

A third simple limit exists when the pump and signal coincide at first, but then the pump walks off. This is most characteristic of stimulated amplification processes (i.e., starting from noise) and may also occur in synchronously pumped systems such as the fiber

Raman amplification soliton laser (FRASL).² The net phase change turns out to be proportional to the integral of the initial pump pulse, and, as Fig. 1(b) shows, the signal spectrum is asymmetric and has wiggles.

We model the general case by considering the integrated phase shift induced on each segment of the signal as the pump walks through. Assume that the

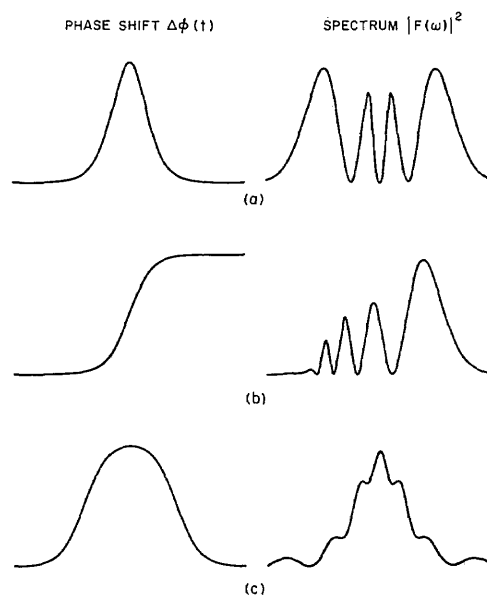


Fig. 1. Phase shifts and spectra corresponding to various degrees of walk-off between pump and signal pulses. (a) Perfect tracking case ($t_0 = \beta = 0$, $2A^2l = 3.5\pi$, $\alpha = 1$); (b) pump and signal coincide initially, and then pump walks off ($t_0 = 0$, $\beta l = 4$, $2A^2/\beta = 3.5\pi$, $\alpha = 1$); and (c) pump walks from trailing edge of signal to the leading edge ($t_0 = -2$, $\beta l = 4$, $2A^2/\beta = 3.5\pi$, $\alpha = 1$).

signal and pump amplitudes are of the form $u_s = \gamma \operatorname{sech}(\alpha t)$ and $u_p = A \operatorname{sech}(t - t_0 - \beta z)$, where t_0 is an initial displacement of the pump, β is proportional to the difference in inverse group velocities, and α scales the width of the signal pulse to the pump pulse width. Here the amplitudes and lengths are normalized with respect to the fundamental soliton,^{5,6} and the effect of dispersion is included only to lowest order through group-velocity mismatch between the pump and signal pulses.¹ The net phase perturbation caused by the pump at an instant t on the signal after traveling a distance l in the fiber is

$$\begin{aligned} \Delta\phi(t) &= 2A^2 \int_0^l \operatorname{sech}^2(t - t_0 - \beta z) dz \\ &= 2 \frac{A^2}{\beta} [\tanh(t - t_0) \\ &\quad - \tanh(t - t_0 - \beta l)] \quad \text{if } (\beta \neq 0). \end{aligned} \quad (1)$$

Neglecting the pulse-shape changes caused by dispersion, the resulting spectrum can be obtained from a Fourier transform⁷ ($\delta\omega$ in normalized units):

$$F(\delta\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} u_s(t) \exp[i\Delta\phi(t)] \exp(-i\delta\omega t) dt. \quad (2)$$

As an example in the use of the above equations, consider the intermediate case in which the pump starts at the trailing edge of the signal and, in the fiber, walks through to the leading edge. We can simulate this by letting $t_0 = -2$ and $\beta l = 4$ in Eq. (1). The Fourier analysis gives the spectrum in Fig. 1(c), which shows that a symmetric spectrum results if the walk-off is symmetric.

Although we have thus far considered equal pump and signal pulse widths, in experiments involving stimulated Raman scattering and fiber Raman lasers, the signal pulse will be narrower than the pump pulse ($\alpha > 1$). In general, as α is increased, the pump power must be increased to obtain similar spectral features of the signal. In other words, the perturbation from XPM is proportional to the net phase variation induced over the width of the signal. For example, we have studied the partial walk-off case of Fig. 1(b) when $\alpha = 5$. Through numerical solutions of Eq. (2), we find that the pump power A^2 must be increased by $\sim \alpha$ to retain the same number of oscillations in the spectrum. On the other hand, for the perfect tracking case shown in Fig. 1(a), increasing A^2 by 5 (for $\alpha = 5$) produces a broad, featureless spectrum. If a narrow signal pulse sits under the peak of a broad pump pulse, it is difficult to maintain the wiggles of Fig. 1(a) because $\partial I_p / \partial t$ is at a minimum.

The FRASL provides a convenient tool for exploring the effects of walk-off because the pump and signal wavelengths are close to the zero-dispersion wavelength λ_0 in the fiber; the pump is in the normal group-velocity dispersion (GVD) regime, whereas the signal is in the anomalous GVD range. The experimental configuration (Fig. 2) consists of a single-fiber FRASL ring cavity into which a narrow-band, tunable étalon may be inserted. A dichroic beam splitter (highly reflective at the pump wavelength λ_p and partially

reflective at the signal wavelength λ_s) couples in the pump and partially couples out the laser. The fiber is single mode, polarization maintaining, and dispersion shifted with a zero of dispersion at $\lambda_0 = 1.536 \mu\text{m}$. The loss is less than 1 dB/km over the wavelength range of interest. To study XPM as a function of walk-off, we use 50-, 100-, and 400-m lengths of fibers. A mode-locked $\text{Ti}^{3+}:\text{KCl}$ color-center laser provides pump pulses of $\tau \sim 10$ psec, which can be tuned between $1.46 < \lambda_p < 1.536 \mu\text{m} = \lambda_0$ (peak power < 200 W in fiber). Because the soliton period^{5,6} Z_0 for the pump is about 25 km, while the longest fiber we use is less than $Z_0/50$, the pump intensity may broaden somewhat, but it does not become flat topped.⁶ With the pump tuned to $\lambda_p = 1.51 \mu\text{m}$, the walk-off length l_{w0} for a signal at $\lambda_s \sim 1.61 \mu\text{m}$ is about 80 m.

Introducing the étalon into the cavity turns the laser into a pump-probe experiment in which we can control the seed fed back into the fiber and observe the spectral broadening in just a single pass. The étalon is surrounded by two beam splitters that enable us to monitor the étalon and fiber outputs. The signals are passed through a bandpass filter to eliminate the pump and then sent to a scanning Fabry-Perot and an autocorrelator.

When a 50-m fiber is used in the FRASL ($l < l_{w0}$), the signal remains with the pump throughout the fiber. With no étalon in the cavity, the signal spectrum is wider than the 300-cm^{-1} free spectral range of our Fabry-Perot. Even with the narrow-pass-band étalon introduced into the cavity, the spectral width remains greater than 300 cm^{-1} [Fig. 3(a)]. Therefore, more or less independent of the seed, the pump in a single pass severely broadens the signal spectrum. As discussed earlier, the spectrum is featureless and without the fringes of Fig. 1(a) because $\alpha \gtrsim 5$ in our experiments with the étalon. Since for our experiments $|\Delta\phi| \sim \alpha 3.5\pi$ (in Fig. 1 we use $|\Delta\phi| = 3.5\pi$), the agreement with Fig. 1 improves when there is more complete walk-off.

If we increase the fiber length to 100 m ($l \sim l_{w0}$), then there is partial walk-off between the pump and signal, and XPM again dominates the spectral features. Without an étalon in the FRASL cavity, the emerging spectrum is wide and has wiggles. By time-dispersion tuning² the FRASL, thus varying the amount of walk-

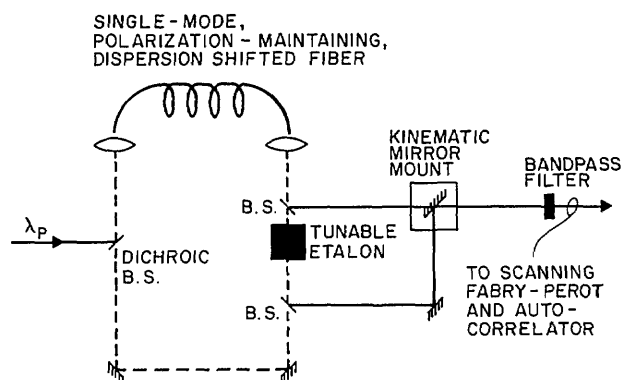


Fig. 2. Modified FRASL (see text). B.S.'s, beam splitters.

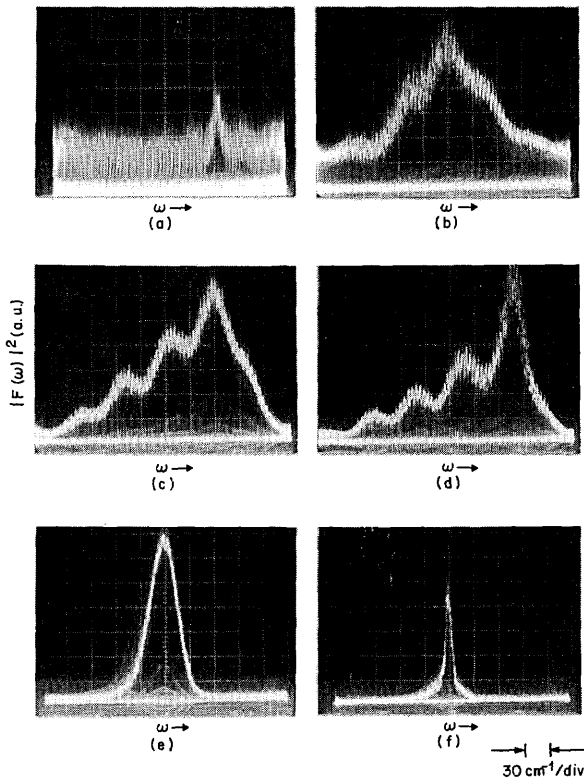


Fig. 3. Experimental spectra for various fiber lengths (l) with and without the tunable étalon in the FRASL cavity. (a) $l = 50$ m with étalon in cavity; (b) $l = 100$ m, no étalon; (c) $l = 100$ m, no étalon, but different FRASL cavity length than in (b); (d) same as (c), except with étalon inserted; (e) $l = 400$ m, no étalon; and (f) same as (e), except with étalon inserted; here, except for the wings, the spectrum is nearly that of the étalon. The vertical scales are in arbitrary units, and the signal strength increases for increasing fiber lengths.

off, the details of the spectrum can be changed. For example, in Figs. 3(b) and 3(c) we show two representative frequency scans that resemble Figs. 1(c) and 1(b), respectively. Even after we insert the étalon into the cavity (and appropriately adjust the cavity length), the spectrum remains qualitatively the same [Fig. 3(d)]. Again, more or less independent of the seed, the pump through XPM broadens and creates wiggles on the signal spectrum in just a single pass.

When there is complete walk-off between pump and signal ($l = 400$ m $\gg l_{w0}$), without an étalon the spectrum is symmetric and secant-hyperboliclike, although still broad [Fig. 3(e)]. The effects of XPM are reduced considerably, but they are not canceled completely because the walk-off is asymmetrized by pump depletion. Furthermore, the large bandwidth of Raman gain does not serve to limit the spectrum significantly. As Fig. 3(f) shows, the addition of the étalon

narrows the spectrum (the narrow peak mimics the seed spectrum). However, XPM still produces a broad spectral feature (at the base of the peak), which is comparable in width with the spectrum without the filter [Fig. 3(e)]. In autocorrelation we find that the low-level wider feature corresponds to a $\tau \sim 250$ fsec peak, while the narrow spectral peak results in a broader $\tau \sim 2.5$ psec pulse.

An ideal amplifier would increase the signal amplitude without distorting the signal's pulse shape or spectrum. But our experiments have shown that XPM makes it difficult to approximate even the ideal amplifier in fibers using Raman or parametric pulse amplification. To be sure, the XPM chirp can be minimized by having complete walk-through of pump and signal pulses in a fiber with little or no loss, gain, and pump depletion, but such conditions would make amplifier inefficient. Moreover, XPM would seem difficult to avoid in stimulated amplification processes: since the noise seed grows under the pump before separating, there can never be complete walk-through of the pump.

An alternative way of minimizing XPM is to use a flat-topped pump, as might result when the pump pulse is propagated down a long fiber with normal GVD.⁶ Since $\Delta\omega \sim |\partial I_p / \partial t|$, the induced modulation can be reduced by keeping the signal under the flat portion of the pump. Computer simulations verify the effectiveness of this technique.

On the other hand, XPM could prove useful in compressing weak signal pulses. For example, by tailoring the walk-off, the signal might be chirped in such a way that it may then be compressed by using a grating pair or an appropriate fiber. This is probably one of the pulse-shaping mechanisms in the FRASL that leads to signal pulses as short as 240 fsec.

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